

Experiment No: 2

Title:

To understand Version Control System / Source Code Management, install Git and create a GitHub account.

Aim:

To study the concept of Version Control System (VCS), install Git, configure user information, and create a local repository for version control operations.

Theory:

A Version Control System (VCS) is a software tool that helps developers manage changes in source code over time. It allows multiple developers to collaborate efficiently while maintaining different versions of files and tracking modifications. Git is a distributed version control system that enables users to work independently and merge their changes later without affecting the main project.

Git provides features such as tracking file history, branching, merging, and collaboration with remote repositories hosted on platforms like GitHub. It improves project reliability, coordination among team members, and supports backup and recovery of previous versions of files.

GitHub is a cloud-based hosting platform that allows developers to store repositories remotely and collaborate with others.

Requirements:

- Git Bash installed on system
 - GitHub account
 - Internet connection
 - Windows/Linux operating system
-

Procedure:

Step 1: Create a directory for version control system workspace

Command:

```
mkdir git-dvcs
```

cd git-dvcs

```
chetna mulki@LAPTOP-284977H7 MINGW64 ~/Desktop/Geet Mulki
$
mkdir git-dvcs
cd git-dvcs
```

Step 2: Check the current working directory

Command:

pwd

```
chetna mulki@LAPTOP-284977H7 MINGW64 ~/Desktop/Geet Mulki/git-dvcs
$ pwd
/c/Users/chetna mulki/Desktop/Geet Mulki/git-dvcs
```

Step 3: Configure Git username globally

Command:

git config --global user.name "geetmulki"

```
chetna mulki@LAPTOP-284977H7 MINGW64 ~/Desktop/Geet Mulki/git-dvcs
$ git config --global user.name "geetmulki"
git config --global user.email "geetmulkiclas@gmail.com"
```

Step 4: Configure Git email globally

Command:

git config --global user.email "geetmulkiclas@gmail.com"

```
chetna mulki@LAPTOP-284977H7 MINGW64 ~/Desktop/Geet Mulki/git-dvcs
$ git config --global --list
user.name=geetmulki
user.email=geetmulki@gmail.com
```

Step 5: Verify configured Git user details

Command:

```
git config --global --list
```

Step 6: Create a new project directory for repository

Command:

```
mkdir git-demo-project
cd git-demo-project
```

```
chetna mulki@LAPTOP-284977H7 MINGW64 ~/Desktop/Geet Mulki/git-dvcs
$ mkdir git-demo-project
cd git-demo-project
```

Step 7: Initialize Git repository

Command:

```
git init
```

```
chetna mulki@LAPTOP-284977H7 MINGW64 ~/Desktop/Geet Mulki/git-dvcs/git-demo-project
$ git init
Initialized empty Git repository in C:/Users/chetna mulki/Desktop/Geet Mulki/git-dvcs/git-demo-project/.git/
```

Result:

Successfully installed and configured Git, created a working directory, and initialized a local repository for version control operations.

Conclusion:

Thus, the Version Control System using Git was studied and implemented successfully by configuring user details and initializing a repository.